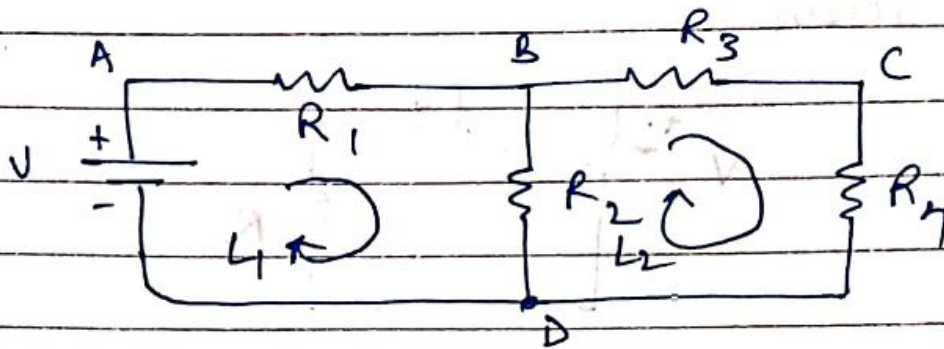


## Basics of Electrical Engg.

Node -

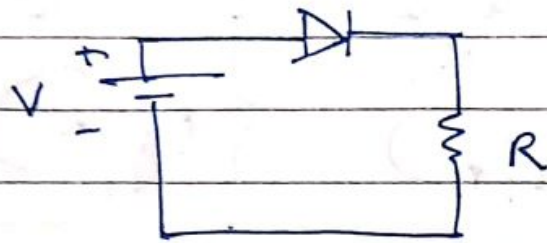


pt. where 2 or more than 2 elements are connected. A, B, C, D are nodes.  
~~A, B~~

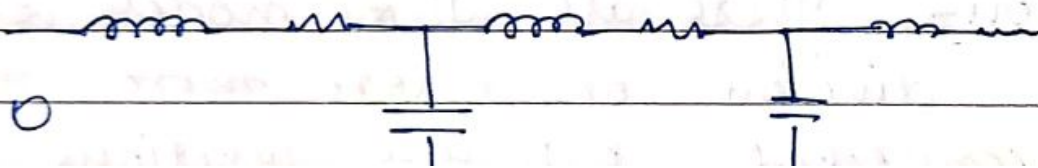
Junction - These are those nodes where there is division or where more than 2 nodes are connected. B, D are junctions.

loop - closed path where no element is encountered more than once.  $L_1$  and  $L_2$  are closed loops. ABCDA is also a closed path.

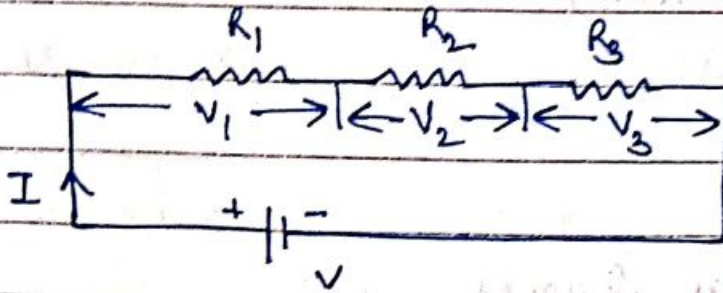
Unilateral circuits are those circuits where the prop. of circuit change with direction of operation



Bilateral circuits are those circuits where prop. do not change with dir. of operations.  
eg:- Transmission lines. Also c/a Reciprocal ckt.





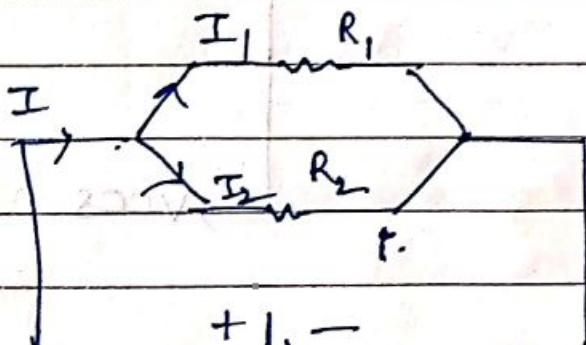
Voltage Divider equation -

$$I = \frac{V}{R_1 + R_2 + R_3}$$

$$V_1 = IR_1 = \frac{VR_1}{(R_1 + R_2 + R_3)}$$

$$V_2 = \frac{VR_2}{R_{eq.}}$$

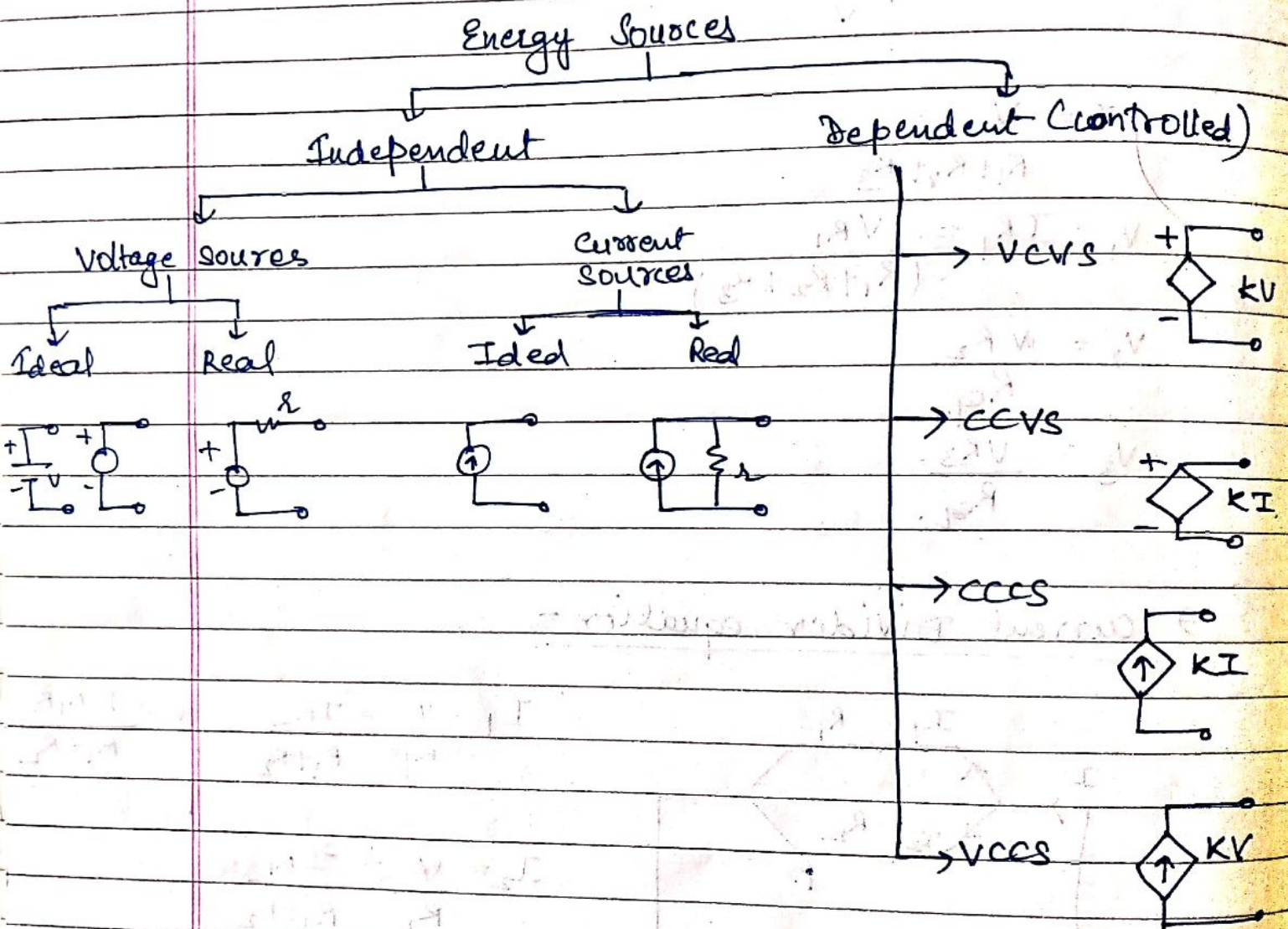
$$V_3 = \frac{VR_3}{R_{eq.}}$$

Current Divider equation -

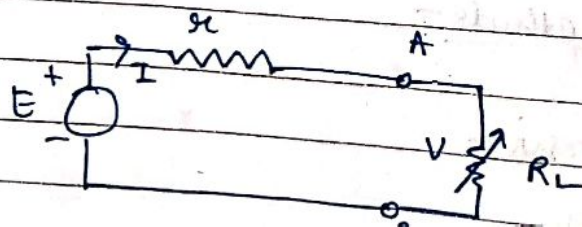
$$I_1 = \frac{V}{R_1} = \frac{IR_2}{R_1 + R_2}$$

$$V = \frac{IR_1 R_2}{R_1 + R_2}$$

$$I_2 = \frac{V}{R_2} = \frac{IR_1}{R_1 + R_2}$$



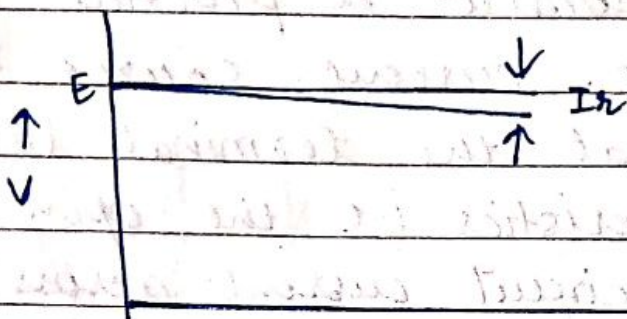
⇒ **Ideal Independent Voltage source -**



$$V = E - Ir$$



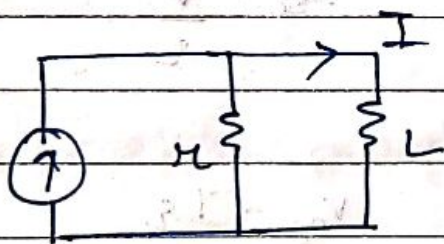
Ideal independent volt. source is a voltage source ~~that~~ ~~is~~ which gives constant voltage across load terminal.



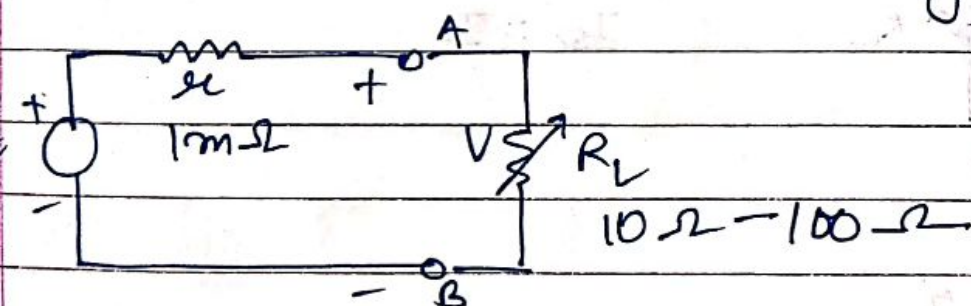
So an ideal vol. source is one which has int. resist. as zero. But ideal is not achieved. So source having a little bit of 'r' is a practical voltage source. But it is kept very low so that 'Ir' is small.

Ideal Independent Current Source: —

Ideal ind. current source is an energy source which gives constant current which is the value of load resistance <sup>ever</sup>.



Internal resist. should be infinite. for making I const. for any L.



$$I_1 = \frac{10}{10^6 + 100}$$

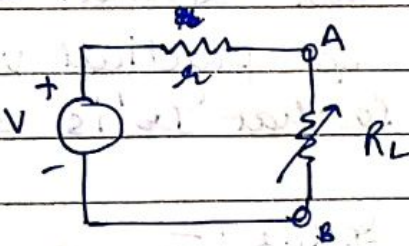
$$T. - 10 =$$



SOURCE CONVERSION:-

Source conversion is a network reduction technique in which we convert a practical voltage source to prac. current source & vice versa. Such that the terminal voltage/ current ampere characteristics i.e. the open circuit voltage & short circuit current across the load terminal remains the same.

Practical Volt. source to Prac. Current source:-

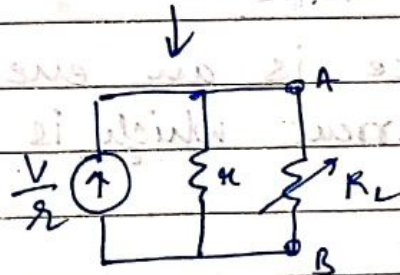


$$V_{oc} = V$$

OC = open circuit

$$I_{sc} = \frac{V}{r}$$

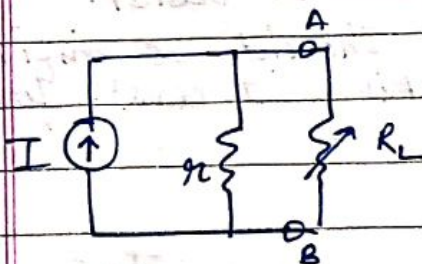
SC = short circuit.



$$V_{oc} = \frac{V}{r} \cdot r = V$$

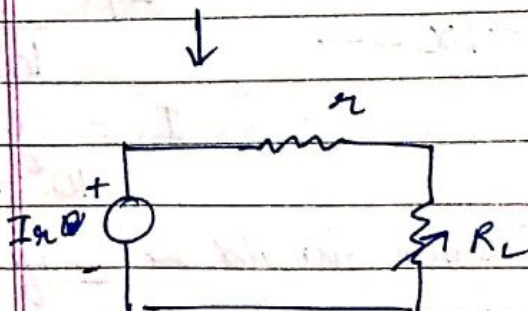
$$I_{sc} = \frac{V}{r}$$

Vice-versa



$$V_{oc} = I r$$

$$I_{sc} = I$$



$$V_{oc} = I r$$

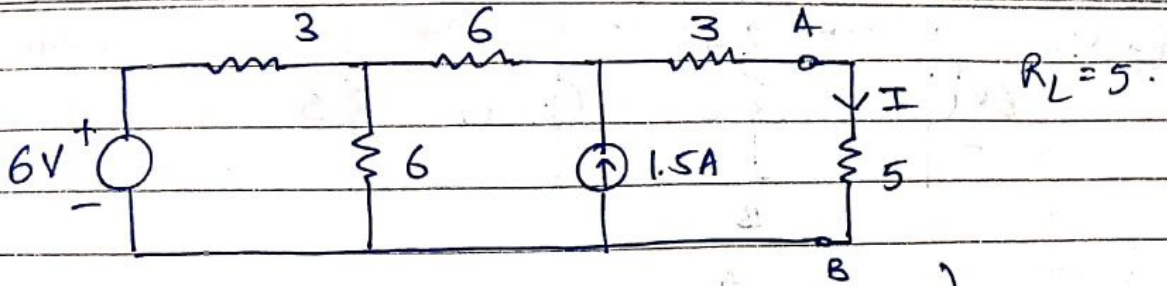
$$I_{sc} = \frac{I r}{r} = I$$



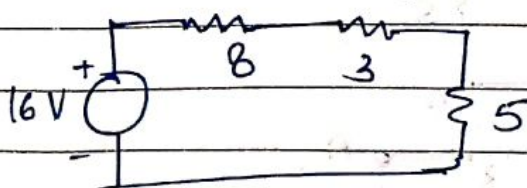
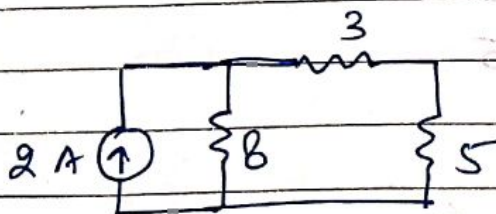
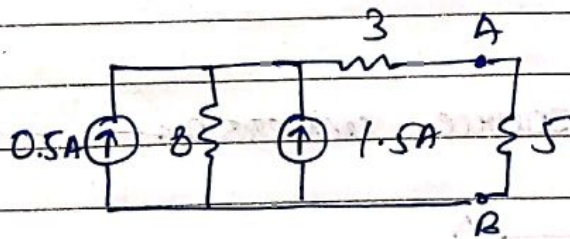
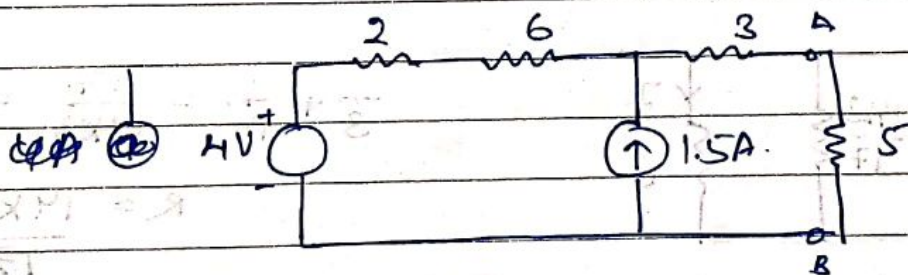
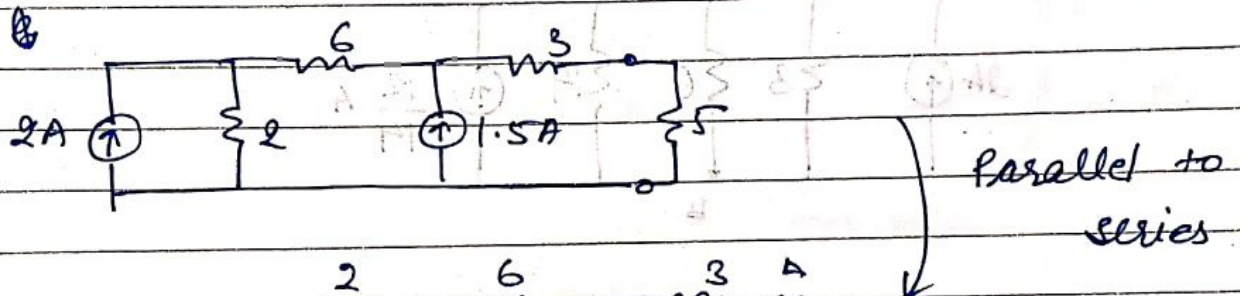
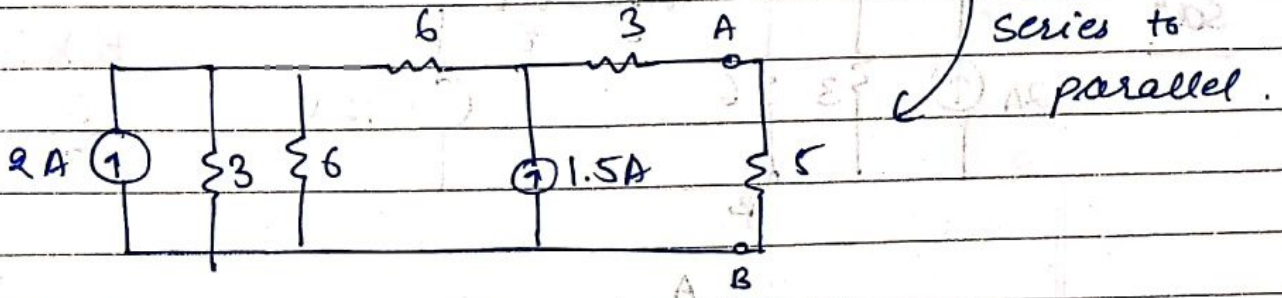
Note - (★) Never combine load resistor with any of the energy sources or other resistors:

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Ques:



Sol:



$$I = \frac{16}{8 + 3 + 5}$$

$$I = 1A$$

- (★) Current sources are added in Parallel
- (★) Voltage sources are added in series.

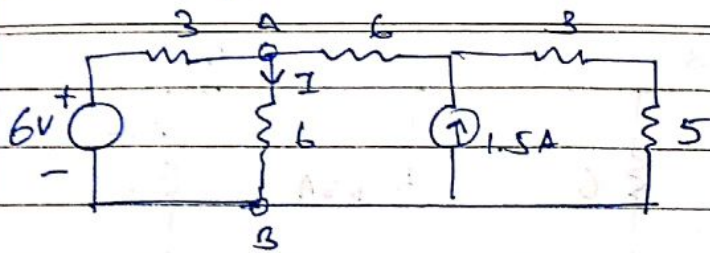


Find I using source conversion:-

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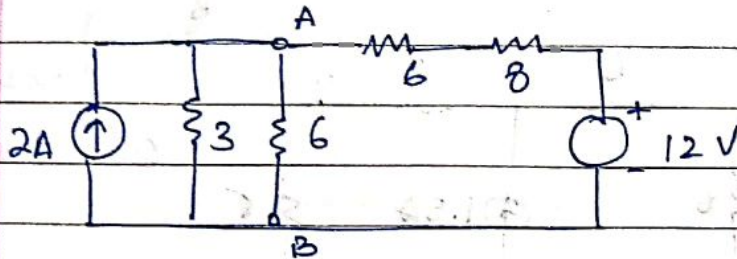
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Ques:



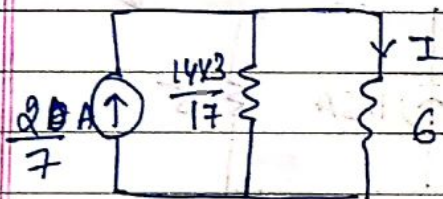
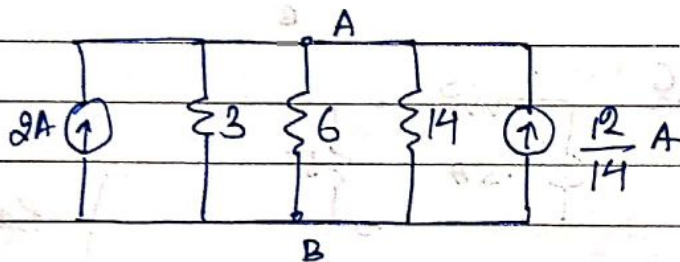
Here  $R_L = 6$   
So we cannot combine it.

Sol<sup>n</sup>:



$$8 \times 1.5 = 12V$$

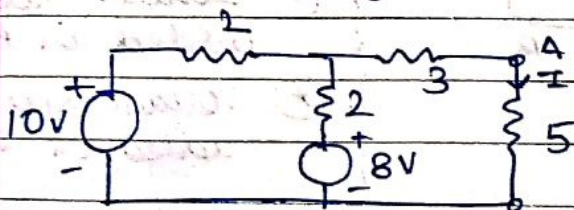
$$\frac{6}{3} = 2A$$



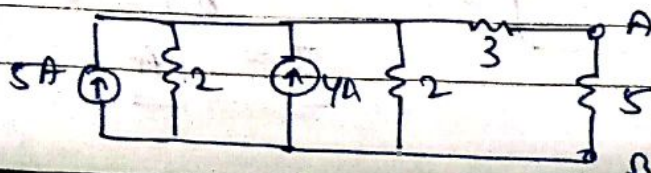
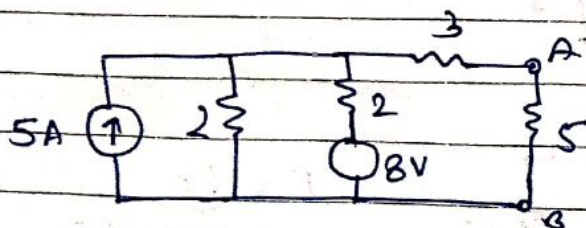
$$\frac{1}{3} + \frac{1}{14} = \frac{17}{14 \times 3} = \frac{1}{R}$$

$$R = \frac{14 \times 3}{17}$$

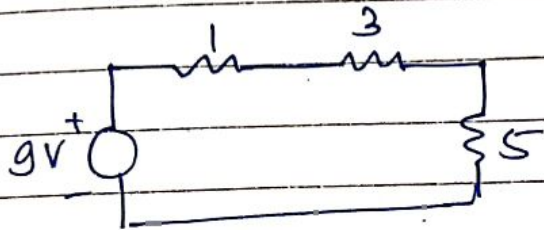
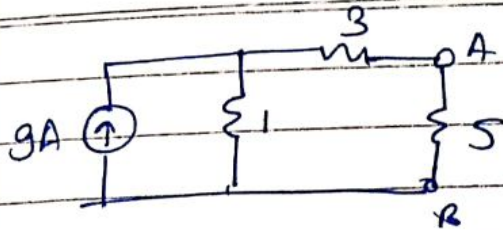
Ques: Find I using source conversion-



Sol<sup>n</sup>:-



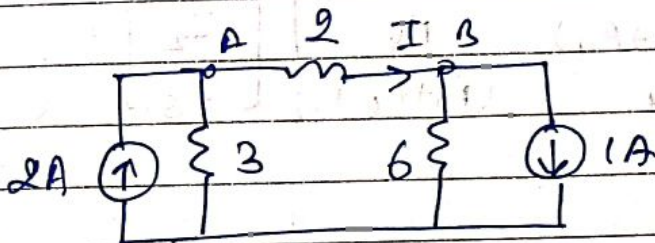




$$\Rightarrow I = \frac{9}{1+3+5}$$

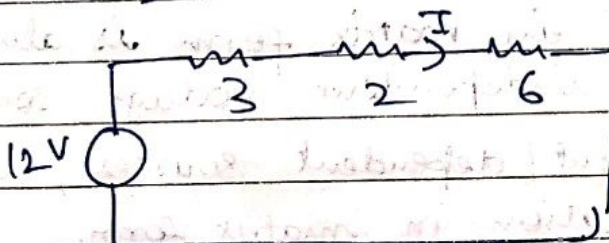
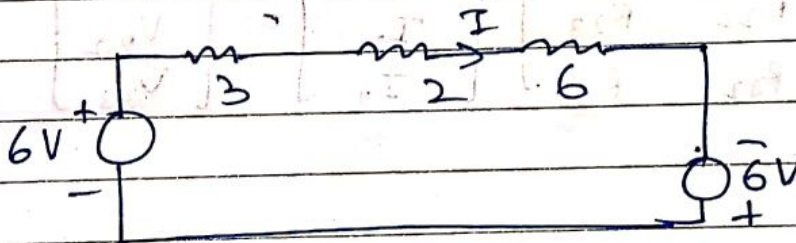
$$I = 1 \text{ A}$$

Ques:



$$R_L = 2$$

④ we cannot directly add 2A & 1A becoz there is resistor in between them. (in series)

Sol<sup>n</sup>

$$I = \frac{12}{3+2+6}$$

$$I = \frac{12}{11} \text{ A}$$

19/01

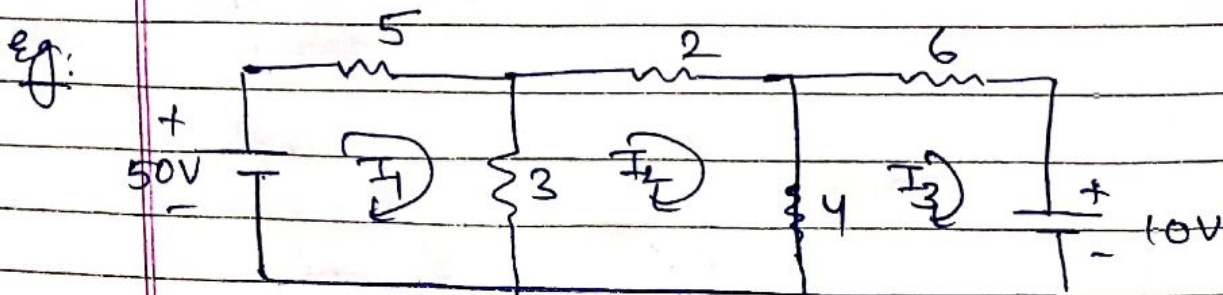
② Loop Matrix Equation:

$$\begin{bmatrix} (R_1 + R_2) & -R_2 & 0 \\ -R_2 & (R_2 + R_3 + R_4) & -R_4 \\ 0 & -R_4 & (R_4 + R_5) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} V_1 - V_2 \\ V_2 \\ V_3 \end{bmatrix}$$

$$\begin{bmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} V_{11} \\ V_{22} \\ V_{33} \end{bmatrix}$$

The mesh equation in matrix form is always form when there are independent voltage sources.

In case of current/dependent sources, we cannot apply mesh equation in matrix form.





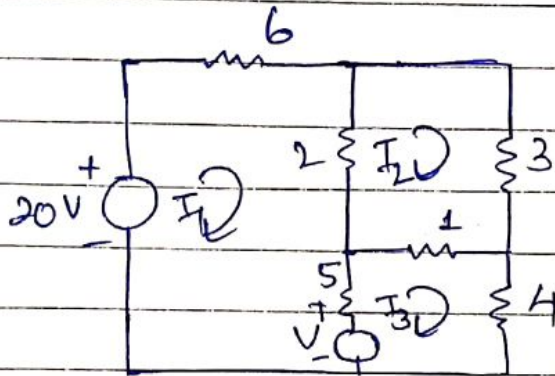
Cramer's Rule  $- I_1 = \frac{\Delta_1}{\Delta}$

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$$\begin{bmatrix} 8 & -3 & 0 \\ -3 & 9 & -4 \\ 0 & -4 & 10 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 50 \\ 0 \\ -10 \end{bmatrix}$$

eg:



Find the value of V  
so that ~~the~~  $I_1 = 0$ .

sol<sup>n</sup>:

$$\begin{bmatrix} 13 & -2 & -5 \\ -2 & 6 & -1 \\ -5 & -1 & 10 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 20-V \\ 0 \\ V \end{bmatrix}$$

becoz at this branch  $I_1$  oppose  $I_2$   
 $I_2$  oppose  $I_3$

Total voltage in loop 1.

$$\begin{bmatrix} 13 & -2 & -5 \\ -2 & 6 & -1 \\ -5 & -1 & 10 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 20-V \\ 0 \\ V \end{bmatrix}$$

$$I_1 = 0 = \frac{\Delta_1}{\Delta}$$

$$\Delta_1 = 0$$

Cramer's formula is more  
easy & efficient over here.

$$\Delta_1 = 0 = \begin{vmatrix} (20-V) & -2 & -5 \\ 0 & 6 & -1 \\ V & -1 & 10 \end{vmatrix} = 0$$

$$(20-V)[60 - 1] + 2(-6V) - 5(-6V) = 0$$

$$1220 - 61V + 2V + 30V = 0$$

$$29V = 1220$$

$$\Rightarrow V =$$

$$1200 - 20 - 53V + 2V + 30V = 0$$

$$1180 - 27V = 0$$

$$27V = 1180$$



⊛ Diagonal elts will be +ve.  
Rest elts will be -ve

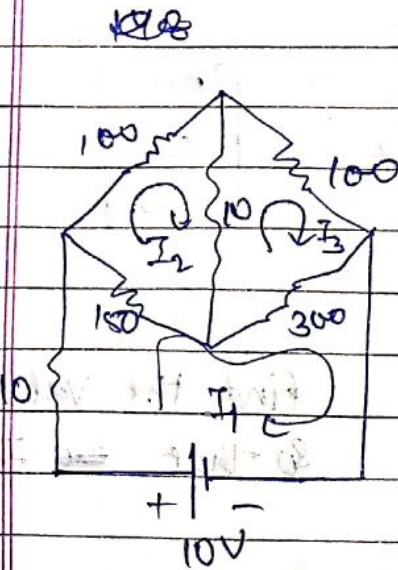
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Q.

Find

using mesh analysis.



Sol<sup>y</sup>

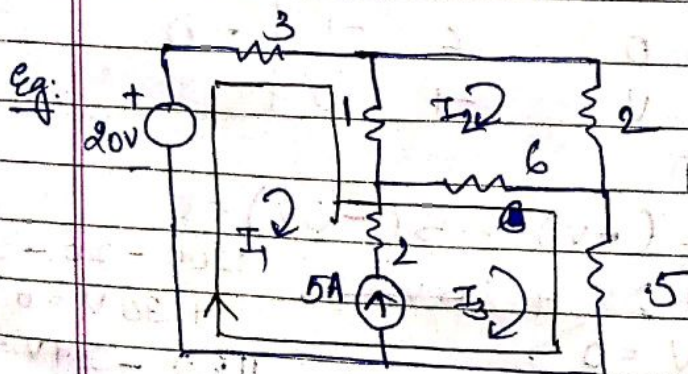
$$\begin{bmatrix} 460 & -150 & -300 \\ -150 & 260 & -10 \\ -300 & -10 & 410 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix}$$

⊛ MESH EQUATIONS WHEN THERE ARE INDEPENDENT CURRENT SOURCES: (SUPER MESH ANALYSIS):-

Can't write in matrix form.

The voltage drop across current source is unknown. So we cannot apply KVL.

In case the cur. source is in a branch which is common to two meshes. So we combine 2 meshes and write KVL.



$$I_3 - I_1 = 5A$$

Loop I<sub>1</sub>

$$20 = 3I_1 - 1(I_1 - I_2)$$

$$-6(I_3 - I_2) - 5I_3 = 0$$

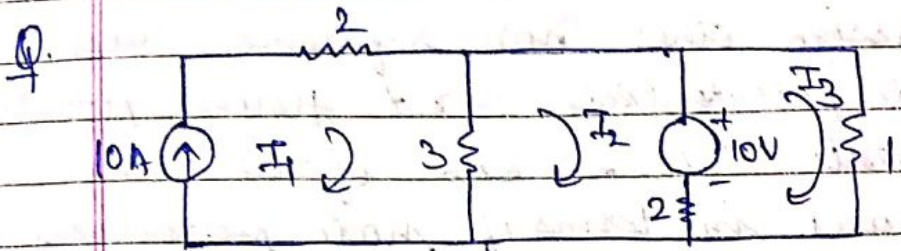
Loop I<sub>2</sub>

$$-1(I_2 - I_1) - 2I_2 - 6(I_2 - I_3) = 0$$

Here we are taking loop I<sub>1</sub> so I<sub>1</sub> - I<sub>3</sub>



NOTE:- Any resistance in series with current source has no meaning.



→ KVL can't be applied in first loop

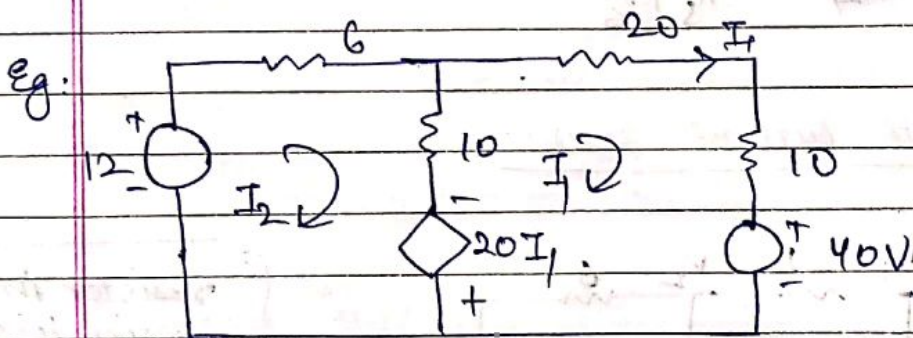
So applying KVL in 2nd & 3rd loop

$$I_1 = 10A$$

$$-3(I_2 - I_1) - 10 - 2(I_2 - I_3) = 0$$

$$-2(I_3 - I_2) + 10 - 1 \times I_3 = 0$$

### (\*) MESH EQUATIONS WITH DEPENDENT SOURCES:-



⇒ In this also we can write in matrix form

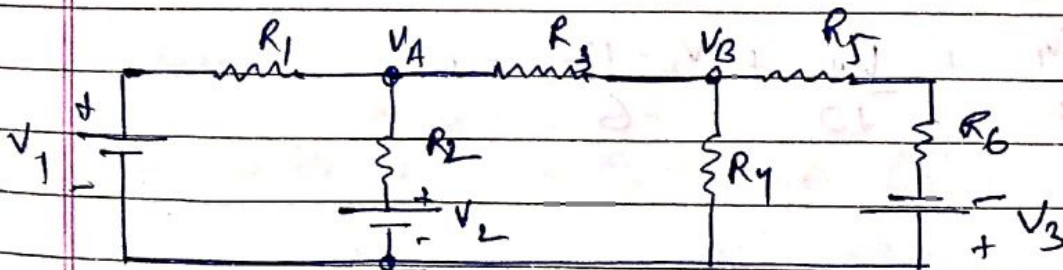
$$12 - 6I_2 - 10(I_2 - I_1) + 20I_1 = 0$$

$$-20I_1 - 10(I_1 - I_2) - 20I_1 - 10I_1 - 40 = 0$$

10/01/20.

### NODAL ANALYSIS:-

① Identify



ground is done at having max. branches

Self node determ. node



If we solve using mesh equat. then we had to write 3 equations. Here we have write 2 only.

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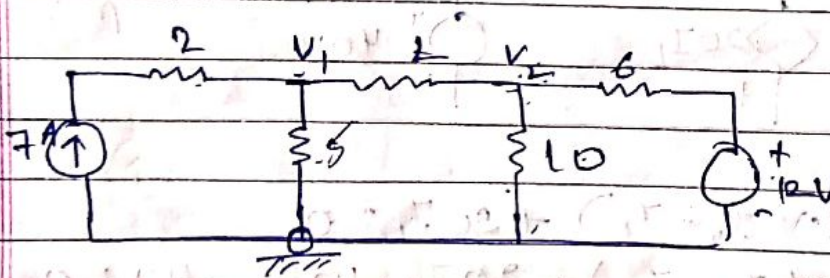
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- ① Identify principle node/jns in the circuit.
- ② Then we assign one of the nodes for the reference node or
- ③ Then we assign other non-reference nodes indep. node potentials v.r.t ground potentials
- ④ Then we write KCL at each of the non-reference nodes in terms of node potentials & the resistances connected to it.

at  $V_A$  
$$\frac{V_A - V_1}{R_1} + \frac{V_A - V_2}{R_2} + \frac{V_A - V_B}{R_3} = 0$$

at  $V_B$  
$$\frac{V_B - V_A}{R_3} + \frac{V_B}{R_4} + \frac{V_B + V_3}{R_5 + R_6} = 0$$

### \* Network with current sources:-



resistor in series is never in calculation for current sources

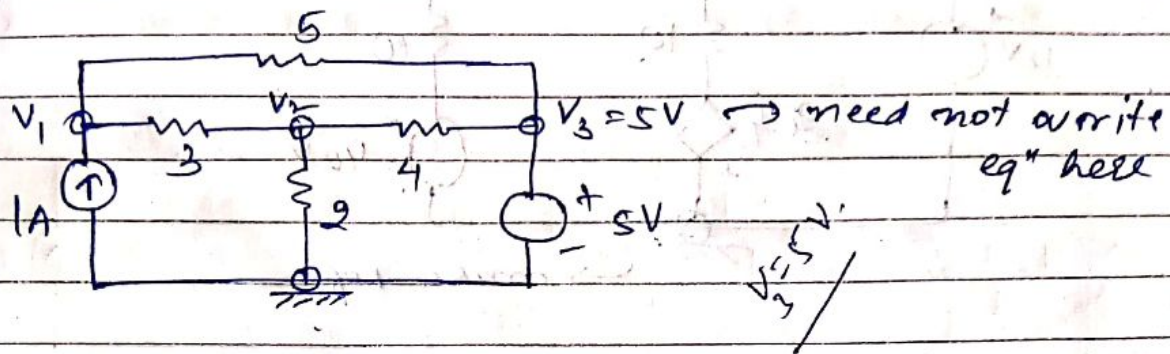
becom we are assum. that current is entering

$$-7 + \frac{V_1}{5} + \frac{V_1 - V_2}{2} = 0$$

$$\frac{V_2 - V_1}{2} + \frac{V_2}{10} + \frac{V_2 - 12}{6} = 0$$

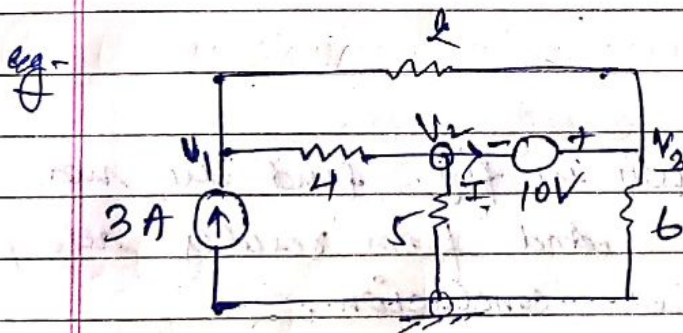


### ⊛ Network with dependent sources:-



$$-1 + \frac{V_1 - V_2}{3} + \frac{V_1 - V_3}{5} = 0$$

$$\frac{V_2 - V_1}{3} + \frac{V_2}{2} + \frac{V_2 - V_3}{4} = 0$$



### Supernode Analysis:-

$$\text{at } V_1, \quad -3 + \frac{V_1 - V_3}{2} + \frac{V_1 - V_2}{4} = 0 \quad (1)$$

$$\text{at } V_2, \quad \frac{V_2 - V_1}{4} + \frac{V_2}{5} + I = 0 \quad (2)$$

$$\text{at } V_3, \quad -I + \frac{V_3}{6} + \frac{V_3 - V_1}{2} = 0 \quad (3)$$

cadding (2) + (3)

$$\frac{V_2 - V_1}{4} + \frac{V_2}{5} + \frac{V_3}{6} + \frac{V_3 - V_1}{2} = 0 \quad (4)$$

$$V_3 = V_2 + 10 \quad (5)$$

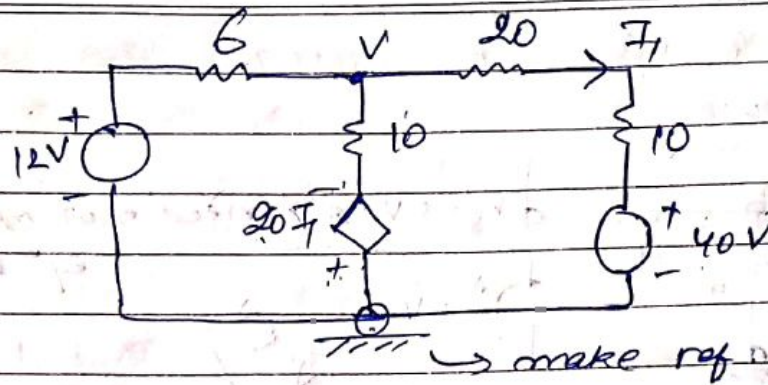
In case of more than 1 ideal voltage source we can't make more than 1 node as ideal node. so, we do supernode analysis

find value of node

potentials & put in (2) or (3) & find value of



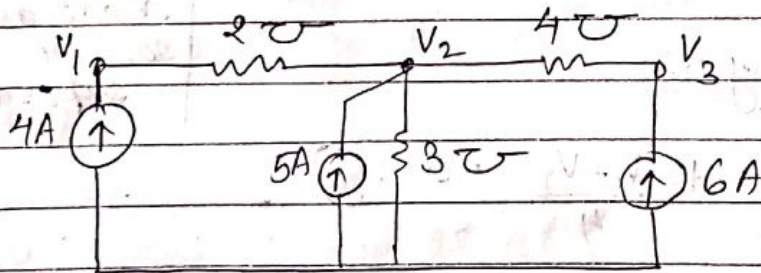
Ques:

making nodal eq<sup>n</sup> -

$$\frac{V-12}{6} + \frac{V-0}{10} + \frac{V-40}{30} = 0$$

$$I_1 = \frac{V-40}{30}$$

Amp Ques: For the circuit shown in fig. find the node voltages  $V_1, V_2, V_3$  and from results attempt to draw generalised conclusion.



If  $V_1, V_2, V_3$  were not assigned then we would have taken  $V_1$  only.

$\Omega = \text{ohm}$

at  $V_1$   $-4 + 2(V_1 - V_2) = 0$

at  $V_3$   $-6 + 4(V_3 - V_2) = 0$

at  $V_2$   $2(V_2 - V_1) + 4(V_2 - V_3) + 3V_2 - 5 = 0$

$$V_1 = 7V, V_2 = 5V, V_3 = 6.5V$$

So, we need not write these

only this is sufficient

also,

$$-4 - 5 - 6 + 3V_2 = 0 \Rightarrow V_2 = 5V$$

Conclusion-

① Any resistance in series with current source is not considered.



# NODE MATRIX EQUATION:-

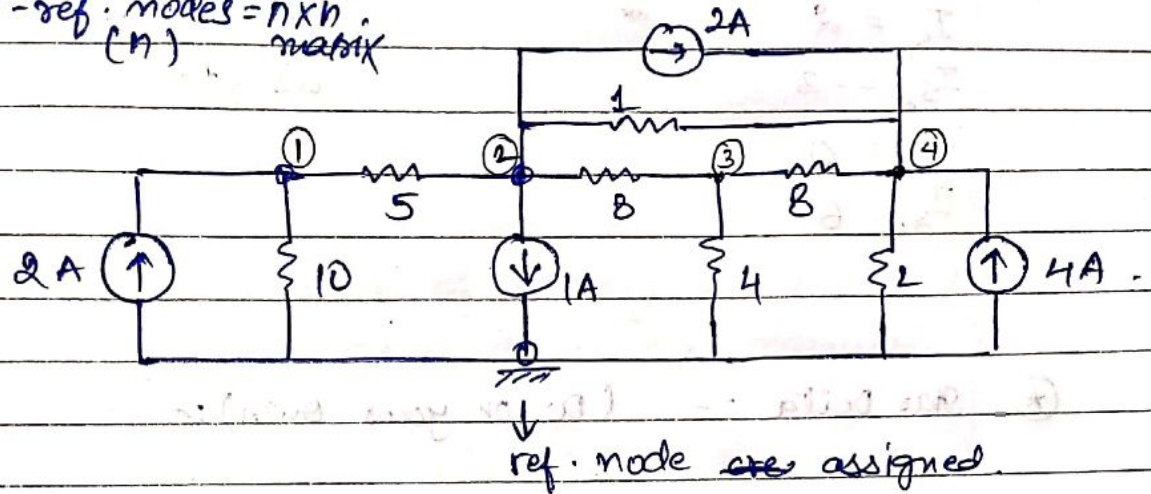
Imp.

Ques:

write node matrix eq<sup>n</sup> for circuit given

[5]

No. of non-ref. nodes =  $n \times n$  matrix  
(n)



$$\begin{bmatrix} G_{11} & G_{12} & G_{13} & G_{14} \\ G_{21} & G_{22} & G_{23} & G_{24} \\ G_{31} & G_{32} & G_{33} & G_{34} \\ G_{41} & G_{42} & G_{43} & G_{44} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \\ V_4 \end{bmatrix} = \begin{bmatrix} I_{S1} \\ I_{S2} \\ I_{S3} \\ I_{S4} \end{bmatrix}$$

$G_{11}$  = Total conductance at node 1

$$G_{11} = 1/10 + 1/5$$

$$G_{22} = 1/5 + 1/8 + 1/1$$

$$G_{33} = 1/8 + 1/4 + 1/2$$

$$G_{44} = 1/8 + 1/1 + 1/2$$

} always +ve. (diagonal elements)

$$G_{12} = G_{21} = -\frac{1}{5} \text{ common resistance}$$

$$G_{13} = G_{31} = 0$$

$$G_{14} = G_{41} = 0$$

$$G_{23} = G_{32} = -\frac{1}{8}$$

$$G_{24} = G_{42} = -\frac{1}{1}$$

$$G_{34} = G_{43} = -\frac{1}{8}$$



if current entering from left, & going to right, then ~~on~~ on right side it ~~is~~ <sup>it's -ve</sup> becomes +ve.

$$I_{S1} = 2$$

$$I_{S2} = -3$$

$$I_{S3} = 0$$

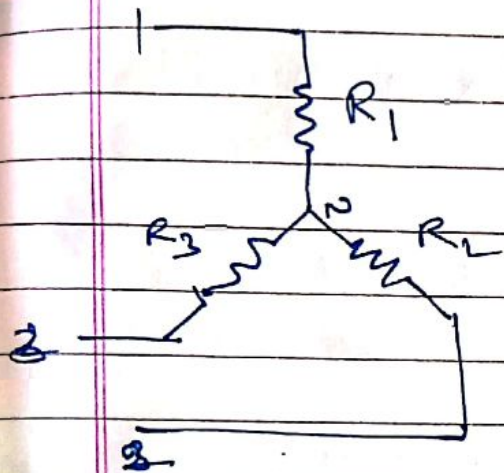
$$I_{S4} = 6$$

on entering from right, it is +ve.

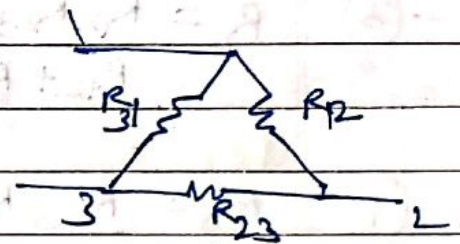
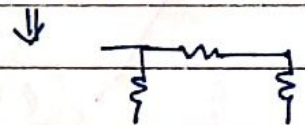
★ Star Delta :- (Do on your own):-

|       |       |       |       |
|-------|-------|-------|-------|
| $V_1$ | $V_2$ | $V_3$ | $V_4$ |
| $V_1$ | $V_2$ | $V_3$ | $V_4$ |
| $V_1$ | $V_2$ | $V_3$ | $V_4$ |
| $V_1$ | $V_2$ | $V_3$ | $V_4$ |

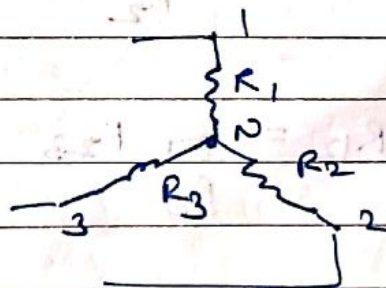
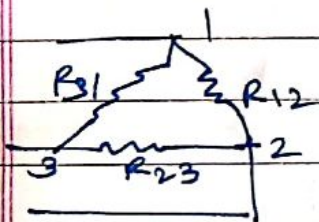


# STAR DELTA TRANSFORMATION:- (Module -1).

Star - Y or T

Δ or Δ  
(Delta)Delta to Star.

For the 2 ccts to be equivalent, the eq. resistance b/w any 2 terminals should be the same.



$$R_1 + R_2 = R_{12} \parallel (R_{23} + R_{31}) = \frac{R_{12}(R_{23} + R_{31})}{R_{12} + R_{23} + R_{31}} \quad \text{--- (1)}$$

$$R_2 + R_3 = \frac{R_{23}(R_{12} + R_{31})}{R_{12} + R_{23} + R_{31}} \quad \text{--- (2)}$$

$$R_1 + R_3 = \frac{R_{31}(R_{12} + R_{23})}{R_{12} + R_{23} + R_{31}} \quad \text{--- (3)}$$

-PTO-



$$2(R_1 + R_2 + R_3) = \frac{R_{12}R_{23} + R_{23}R_{31} + R_{31}R_{12}}{R_{12} + R_{23} + R_{31}} \quad (4)$$

$$(4) - (2) \quad R_1 = \frac{R_{12}R_{31}}{R_{12} + R_{23} + R_{31}} \quad (5)$$

$$(4) - (3) \quad R_2 = \frac{R_{12}R_{23}}{R_{12} + R_{23} + R_{31}} \quad (6)$$

$$(4) - (1) \quad R_3 = \frac{R_{23}R_{31}}{R_{12} + R_{23} + R_{31}} \quad (7)$$

$\Rightarrow$  If the 3 resistances are equal, then equivalent star value is  $\frac{1}{3}$  times the delta value.  
Star to Delta.

$$\frac{(5)}{(6)} \quad \frac{R_1}{R_2} = \frac{R_{31}}{R_{23}} \Rightarrow R_{31} = \frac{R_1}{R_2} R_{23}$$

$$\frac{(5)}{(7)} \quad \frac{R_1}{R_3} = \frac{R_{12}}{R_{23}} \Rightarrow R_{12} = \frac{R_1}{R_3} R_{23}$$

$$R_1' = \frac{R_1}{R_3} \cdot R_{23} \cdot \frac{R_1}{R_2} \cdot R_{23}$$

$$\frac{R_1 \cdot R_{23} + R_{23} + \frac{R_1}{R_2} R_{23}}{R_2}$$

$$\Rightarrow R_{23} = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1}$$

$$\rightarrow R_{23} = R_2 + R_3 + \frac{R_2 R_3}{R_1}$$

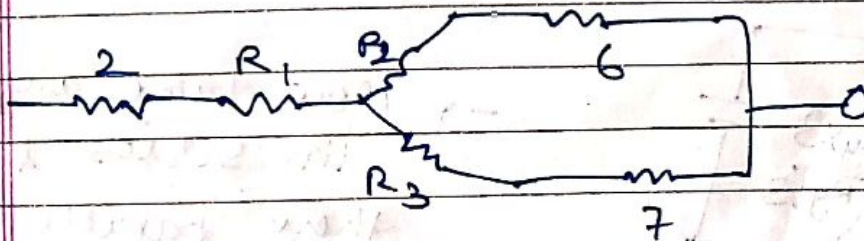
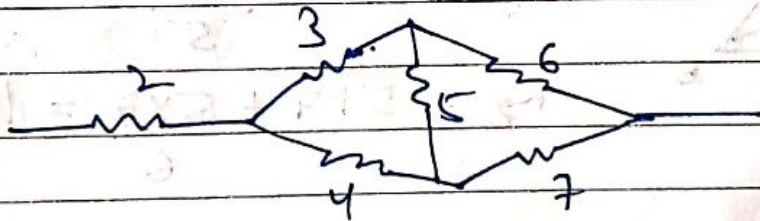
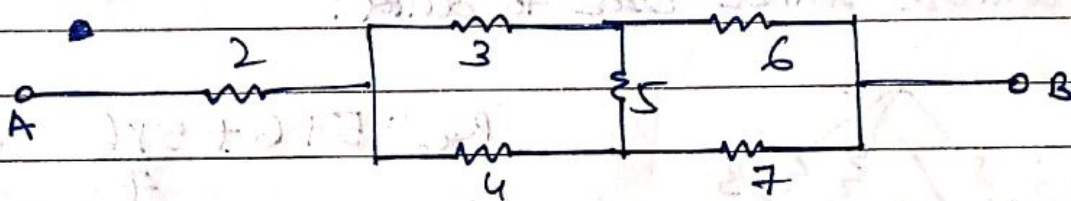


By symmetry,

$$\left. \begin{aligned} \rightarrow R_{12} &= R_1 + R_2 + \frac{R_1 R_2}{R_3} \\ \rightarrow R_{31} &= R_1 + R_3 + \frac{R_1 R_3}{R_2} \end{aligned} \right\}$$

$\Rightarrow$  If the 3 resistances are equal, the equivalent delta value is 3 times the star value.

Ques:

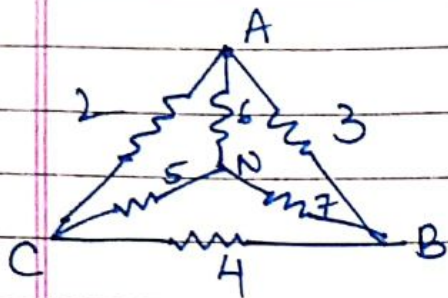


$$R_1 = \frac{3 \times 4}{3+4+5}, \quad R_2 = \frac{3 \times 5}{3+4+5}$$

$$R_3 = \frac{4 \times 5}{4+5+3}$$



Ques. Find  $R_{AB}$  +  $R_{AN}$ .

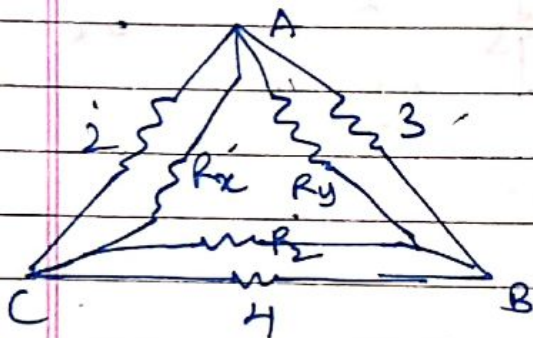


$$R_{AB} = ?$$

$$R_{AN} = ?$$

For  $R_{AB}$ .

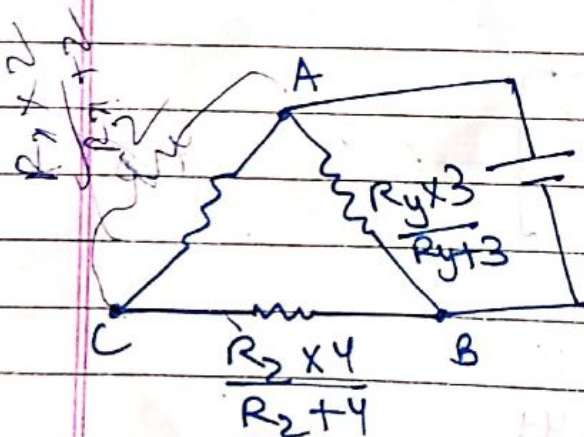
convert inner star to delta.



$$R_x = 5 + 6 + \frac{5 \times 6}{7} = 15.28$$

$$R_y = 6 + 7 + \frac{6 \times 7}{5} = 21.4$$

$$R_z = 5 + 7 + \frac{5 \times 7}{6} = 17.83$$



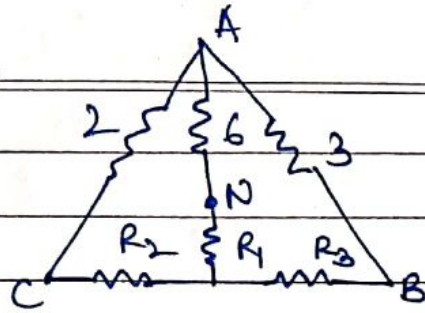
→ Now take any 2 in series & then parallel with the 3rd.

For finding  $R_{AN}$  do not convert inner star to delta becoz on converting we will lose point N.



POT RAN.

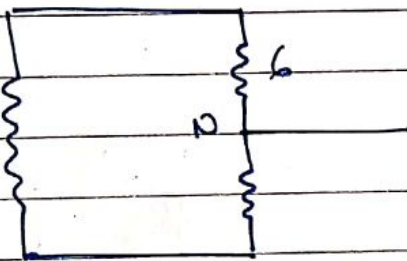
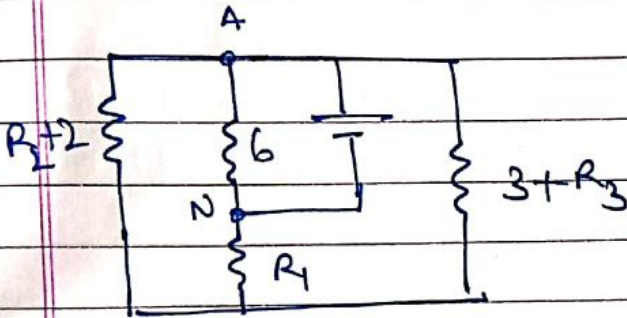
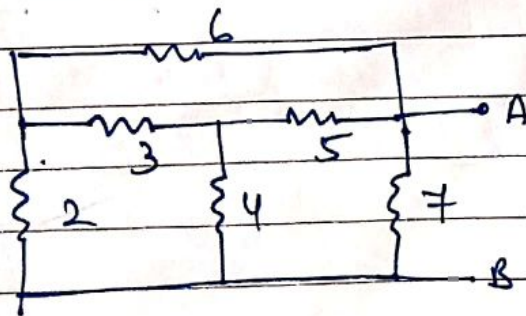
Q



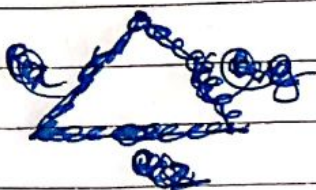
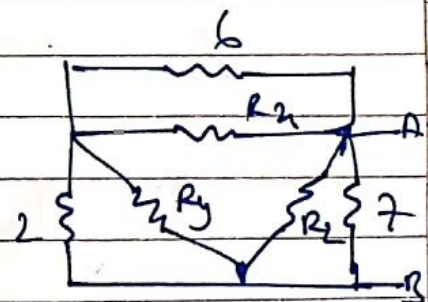
$$R_1 = \frac{5 \times 7}{4 + 5 + 7}$$

$$R_2 = \frac{4 \times 5}{4 + 5 + 7}$$

$$R_3 = \frac{4 \times 7}{4 + 5 + 7}$$

Ques:- Find  $R_{AB}$ .

=



$$R_x = 3 + 5 + \frac{3 \times 5}{4}$$

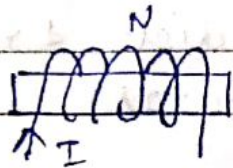
$$R_y = 3 + 4 + \frac{3 \times 4}{5}$$

$$R_z = 4 + 5 + \frac{4 \times 5}{3}$$



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## (★) MAGNETIC CIRCUITS:- (Imp.)



The magnetic flux lines always forms closed loops & closed path followed by it is c/a magnetic circuit.

Magnetomotive force (MMF): (It produces flux)

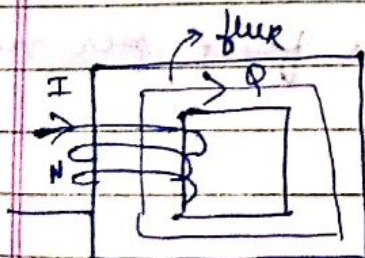
$$F = IN \quad \text{Amperturns (AT)} \\ \text{Amps.}$$

Magnetic field intensity  $\hat{=}$  is MMF per unit length of magnetic flux path.

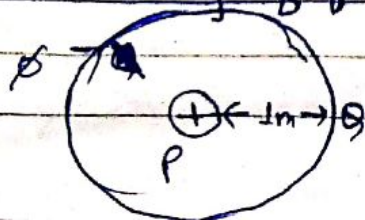
$$H = \frac{\text{MMF}}{\text{mean length of mag. circuit path}}$$

|                    |                                        |
|--------------------|----------------------------------------|
| $H = \frac{IN}{l}$ | $\frac{\text{AT}}{\text{m.}}$<br>unit. |
|--------------------|----------------------------------------|

 $\Rightarrow \boxed{\text{MMF} = Hl}$



Permeability of free space - long st. line in vacuum carrying current of 1 amp.





let us suppose that the return conductor is at large dist so it does not effect the mag. field. Direction of current is clockwise. length of mag. path at a dist-  $1\text{m}$  at  $P$  is  $2\pi r$  i.e  $2\pi$ . ( $l = 2\pi r = 2\pi$ )

So MMF producing the flux in magnetic circuit is given by  $\text{MMF} = IN$   
 $= 1 \times 1$   
 $\text{MMF} = 1 \text{ AT}$

$$H = \frac{IN}{l} = \frac{1}{2\pi} \text{ AT/m}$$

Now consider another long st. conductor carrying current  $I$  situated at a dist. of  $1\text{m}$  from it. The force of conductor  $Q$  due to magnetic flux is given by

$$\text{MMF} = IN = 1 \times 1 = 1 \text{ AT}$$

$$H = \frac{IN}{2} = \frac{1}{2\pi} \text{ AT/m}$$

• From the def. of Ampere, the force per metre at  $Q$  is  $2 \times 10^{-7}$

Force on conductor. at  $Q = BIl = B \times 1 \times 1$

$$Q = B = 2 \times 10^{-7} \text{ Newton}$$



$$\mu_0 = \frac{B}{H} = \frac{2 \times 10^{-7}}{1/2\pi} = 4\pi \times 10^{-7} \frac{\text{Wb/m}^2}{\text{A/m}} = \frac{\text{Wb}}{\text{Am}} \left( \frac{\text{N}}{\text{m}} \right)$$

### \* RELATIVE PERMEABILITY ( $\mu_r$ ) :-

When MMF applied to ferro magnetic material the flux produced is v. large as produced in vacuum of non mag. material.

The ratio of flux<sup>density</sup> produced in a material with that of flux dens. produced in air or non-magnetic material.

$$\mu_r = \frac{\text{flux density in FM material}}{\text{flux density in air or vacuum}}$$

for air  $\mu_r = 1$ .

$\mu_r \mu_0 = \mu$  is the absolute permeability.

Permeability is the ability of a material for the flux to permeate. It is the measure of ease with which material can be magnetised.

$\mu_r$  is not constant. It changes with diff. values of MMF. With nickel, iron alloys  $\mu_r$  can be as high as  $2 \times 10^5$ .

### (\*) RELUCTANCE :- (S)

The opposition offered by ~~the~~ a mag. ckt to the est. of mag. flux is c/a reluctance of magnetic circuit.



→ Imp: Analogy b/w electrical & mag. circuits  
 → flux analogous to current in elec. circ.

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- Directly prop. to length.  
 Inversely prop. to area of cross section of mag. circuit path.

$$S \propto \frac{l}{a} \Rightarrow \boxed{S = \frac{l}{\mu_0 \mu_r a}}$$

$l$  is the length of mag. circuit path.  
 $a$  is the area of cross section of mag. circuit path.

$$\text{MMF} = Hl = \frac{B}{\mu_0 \mu_r} l = \frac{\phi l}{\mu_0 \mu_r a}$$

$$\boxed{\text{MMF} = S\phi}$$

$$S = \frac{\text{MMF}}{\phi} \quad \text{A/Wb}$$

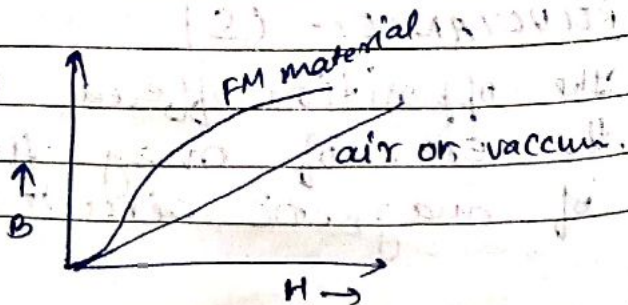
- Reciprocal of reluctance is called permeance.

$$\text{Permeance} = \frac{1}{S} \quad \text{Wb/A (H)}$$

- ★ Force b/w Parallel Conductors: (Do from Book)  
 Derivation & Numerical (Imp.)

$$F_c = 2 \times 10^{-7} \frac{I_1 I_2 l}{d}$$

- ★ Magnetization Curve :-  
 or BH Curve





## ★ LAWS OF GOVERNING MAGNETIC CIRCUITS:-

⇒ There is analogy b/w magnetic & DC electrical circuit.

| Magnetic ckt.            | DC resistive ckt          |
|--------------------------|---------------------------|
| mmf (AT)                 | emf (V)                   |
| H AT/m                   | elec. field intensity V/m |
| $\phi$ Wb                | Current (A)               |
| (Reluctance) $S$ A/Wb    | Resistance ( $\Omega$ )   |
| Permeance $\frac{Wb}{A}$ | Conductance ( $\sigma$ )  |
| $\mu$                    | $(\frac{1}{\rho})$        |

$$mmf = S\phi$$

$$S = \frac{1}{\mu_0 \mu_r} \frac{l}{a}$$

$$emf = RI$$

$$R = \frac{\rho l}{a}$$

## ★ DISTINCTION B/W ELEC. & MAG. CKTS -

1. Electric current is actual flow of  $e^-$  but magnetic flux lines are imaginary.
2. In ferro. mag. materials permeability varies with flux density.  
conductivity is independent of current density within normal operating range.
3. There are good elec. insulators so that the current can be confined to flow in the defined path but all the mag. flux cannot be confined prescribed path. There is always some leakage flux.
4. Flow of elec. current involves continuous expenditure of energy, bt. in mag. ckt energy req. to produce flux bt. no energy req. to maintain it.



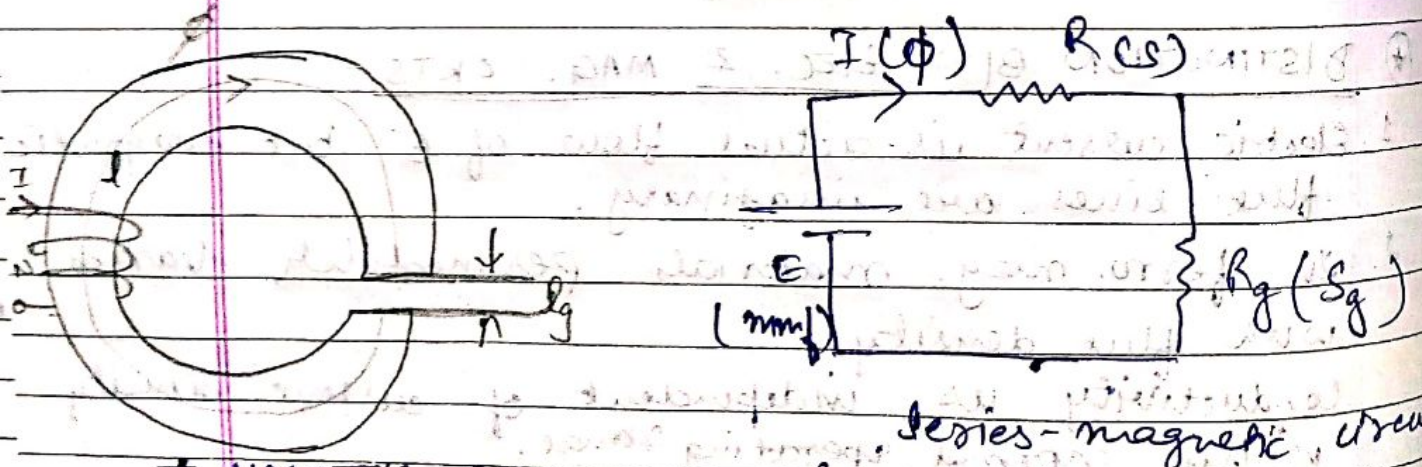
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# fringing :- To make B-H graph of ferro magnetic material more linear

short  
does  
come  
not  
normal

Generally small air gaps are provided in mag. circuits to make B-H curve more linear. When flux lines cross the air gap. They tend to bulge out of edges of core. Which makes effective area of cross section more than that of core. Hence, flux density is less. This effect is fringing.

If air gap v. small then fringing can be neglected. And to take into a/c of fringing effect, empirical factor is introduced.



# MAGNETIC CKT CALCULATIONS:-

→ Series Mag. Circuit:-

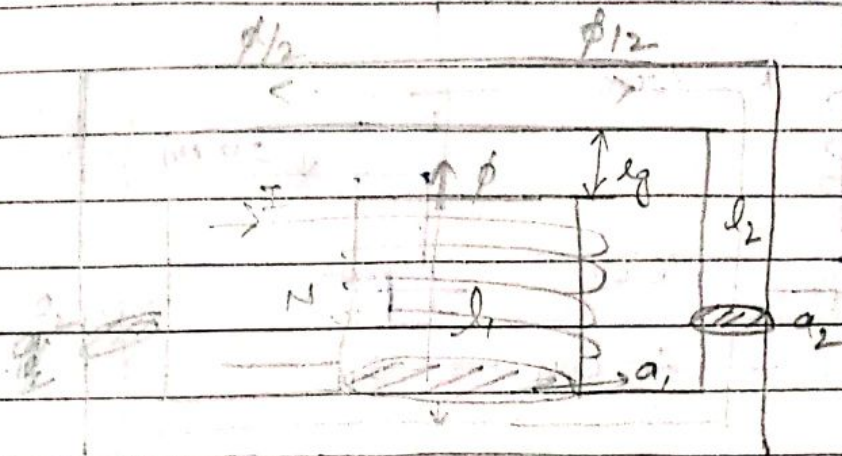
$$E = I(R + R_g)$$

$$\text{mmf} = \phi (l + l_g)$$

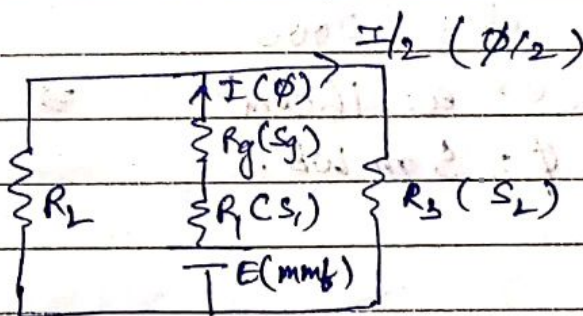
$$= \phi \left( \frac{l}{\mu_0 \mu_r a} + \frac{l_g}{\mu_0 a} \right)$$



→ Parallel Magnetic ckt:-



Equivalent  
electrical  
circuit



$$E = I(R_1 + R_g) + \frac{I}{2}(R_2)$$

$$\text{mmf} = \Phi(S_1 + S_g) + \frac{\Phi}{2}S_2$$

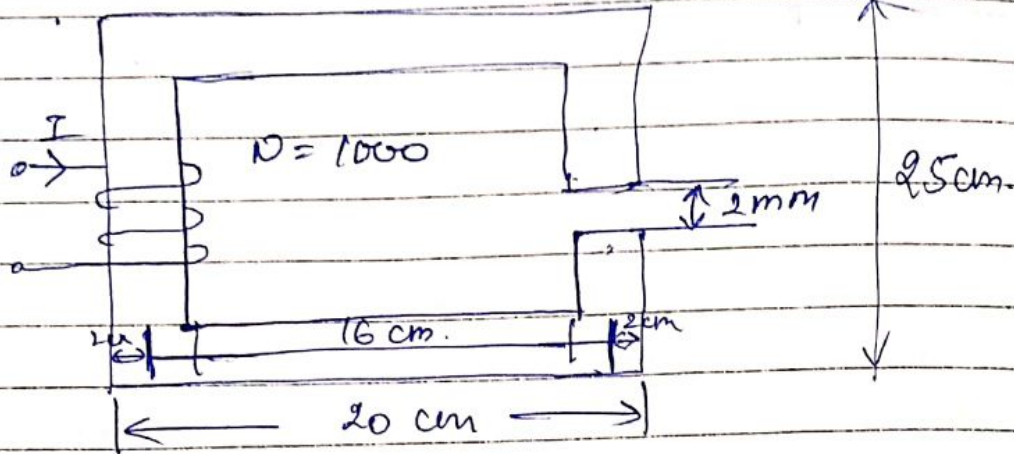
$$S_1 = \frac{1}{\mu_0 \mu_r} \frac{l_1}{a_1}$$

$$S_2 = \frac{1}{\mu_0 \mu_r} \frac{l_2}{a_2}$$

$$S_g = \frac{1}{\mu_0} \frac{l_g}{a_g}$$



Q.



$$\mu_r = 2000$$

$$\text{sq. cross section } \Rightarrow a = 16 \text{ cm}^2$$

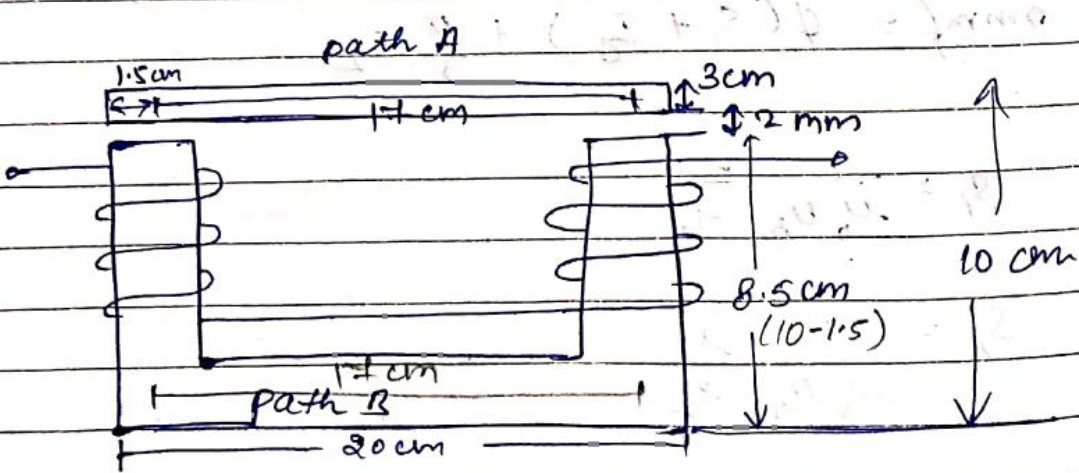
$$\phi = 5 \text{ mwb.}$$

length of one side = 4 cm.

study on own. Q.

What is hysteresis? Define cohesive force & relative flux density. Draw hysteresis loop.

Q.



The mag. frame is made of sq. cross section of 3 cm side. air gap is 2 mm wide. exciting current = 1 A.

$$\mu_r = 1000 \text{ Path A}$$

$$1200 \text{ Path B}$$

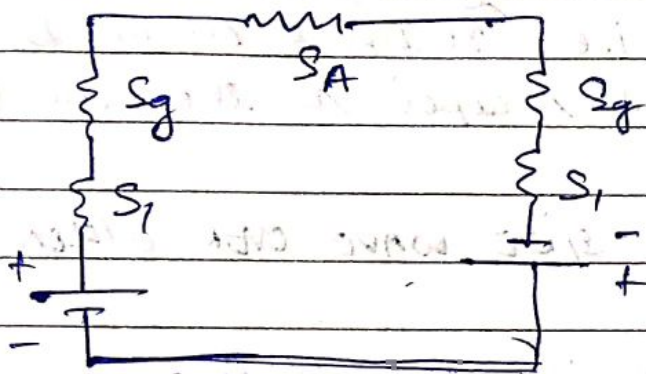
Calculate reluctance of path A & path B



If current dir. same then dir. of flux also same.

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$$S_g + S_1 = S_B$$

Q. A ring has a dia of 21 cm & cross section of 10 cm<sup>2</sup>. The ring is made of semi circular sections of cast iron & cast steel with each <sup>joint</sup> having reluctance of air gap of 2 mm. Find mmf req. to produce the flux of  $8 \times 10^{-8}$  wb.

$\mu_r$  of cast steel = 88.

$\mu_r$  of cast iron = 166.